MSR Dataset Schema

Four CSV files, linked by `issue_id` (`repo_name#issue_number`).

issues.csv

5,000 rows. One row per issue, combining GH Archive data with live GitHub API data.

Column	Type	Description
`issue_id`	string	Join key, `repo_name#issue_number`
`repo_name`	string	Repository, `owner/name`
`issue_number`	int	Issue number within the repo
`title`	string	Issue title from GH Archive
`body`	string	Issue body from GH Archive
`state_final`	string	Last state seen in the GH Archive event stream, open or closed. This can be outdated since the archive only covers a snapshot in time, `api_state` below reflects the current state
`author_login`	string	Issue author login, for example sweep-ai[bot], coderabbitai[bot], or a regular GitHub username
`agent_type`	string	AI coding agent label (automatic guess), for example CodeRabbit, Devin, Claude_Code
`n_gharchive_events`	int	Number of GH Archive events aggregated into this issue. A single issue can appear in the archive multiple times, for example once when opened, again when labeled, again when closed, this column counts how many such events were combined into this one row
`comments_count_reported`	float	GH Archive self-reported comment count, always empty in this export
`created_at`	datetime	Earliest GH Archive event timestamp (UTC)

Column	Type	Description
`closed_at`	datetime	Latest closed-action timestamp in the archive stream (UTC), empty if never closed in the stream
`time_to_close_hours`	float	Hours from `created_at` to `closed_at`, empty if open
`html_url`	string	GitHub issue URL
`api_state`	string	Live GitHub state, open or closed
`state_reason`	string	GitHub close reason, for example completed, not planned, duplicate
`locked`	bool	Whether the conversation is locked
`draft`	bool	Draft flag, mostly relevant for PRs
`is_pull_request`	bool	True if this GitHub issue is actually a pull request
`author_type`	string	GitHub account type of the author, User or Bot
`api_created_at`	datetime	Created timestamp from the GitHub API (UTC)
`api_updated_at`	datetime	Last updated timestamp from the GitHub API (UTC)
`api_closed_at`	datetime	Closed timestamp from the GitHub API (UTC), empty if open
`comments_api`	int	Comment count reported by the GitHub API
`milestone_title`	string	Milestone title if assigned
`milestone_state`	string	Milestone state if assigned, open or closed
`reactions_total`	int	Total reaction count
`reactions_plus_one`	int	Count of +1 reactions
`reactions_minus_one`	int	Count of -1 reactions
`reactions_laugh`	int	Count of laugh reactions

Column	Type	Description
`reactions_hooray`	int	Count of hooray reactions
`reactions_confused`	int	Count of confused reactions
`reactions_heart`	int	Count of heart reactions
`reactions_rocket`	int	Count of rocket reactions
`reactions_eyes`	int	Count of eyes reactions
`n_labels`	int	Number of labels on the issue
`n_assignees`	int	Number of assignees on the issue
`labels`	string	Semicolon-joined label names, for example bug;enhancement;good first issue
`n_comments_actual`	int	Number of rows in `issue_comments.csv` linked to this issue

issue_comments.csv

1,468 rows. One row per comment on an issue in `issues.csv`.

Column	Type	Description
`comment_id`	int	GitHub comment id
`issue_id`	string	Foreign key to `issues.csv`
`issue_number`	int	Issue number within the repo
`repo_name`	string	Repository, `owner/name`
`body`	string	Comment text
`author_login`	string	Comment author login
`agent_type`	string	AI coding agent label on the comment event
`created_at`	datetime	Comment timestamp (UTC)

issue_labels.csv

4,764 rows. One row per issue-label pair.

Column	Type	Description
`issue_id`	string	Foreign key to `issues.csv`
`label`	string	A single label name

issue_assignees.csv

2,041 rows. One row per issue-assignee pair.

Column	Type	Description
`issue_id`	string	Foreign key to `issues.csv`
`assignee_login`	string	GitHub login of an assignee

Notes

`agent_type` is an automatic guess, based on signals like bot account names and text patterns, not confirmed fact.

Not every issue has comments, labels, or assignees. Most issues have none of these.